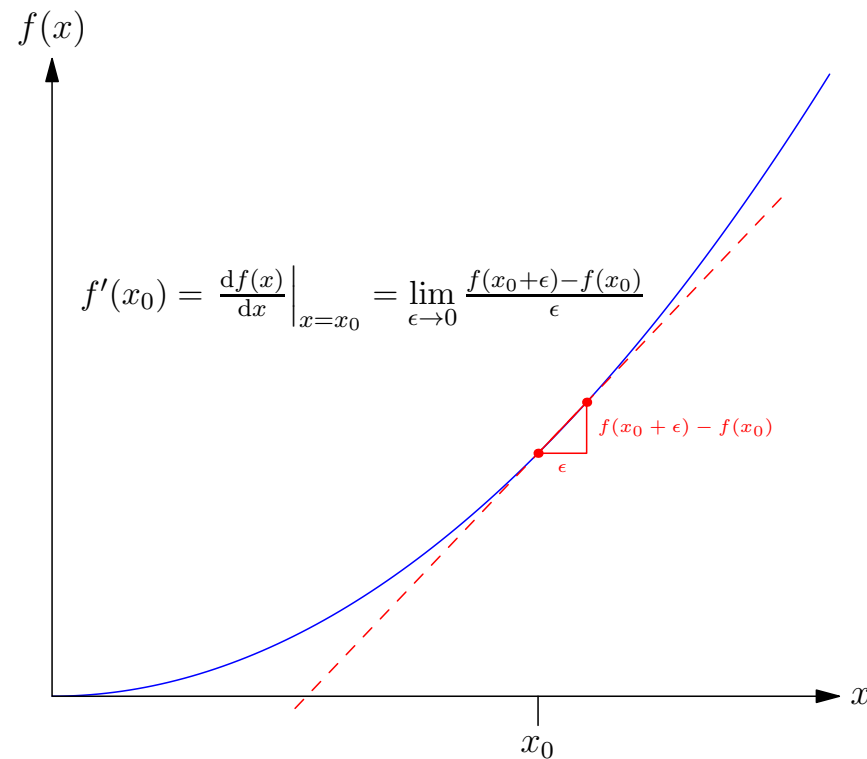


Mathematics of Machine Learning

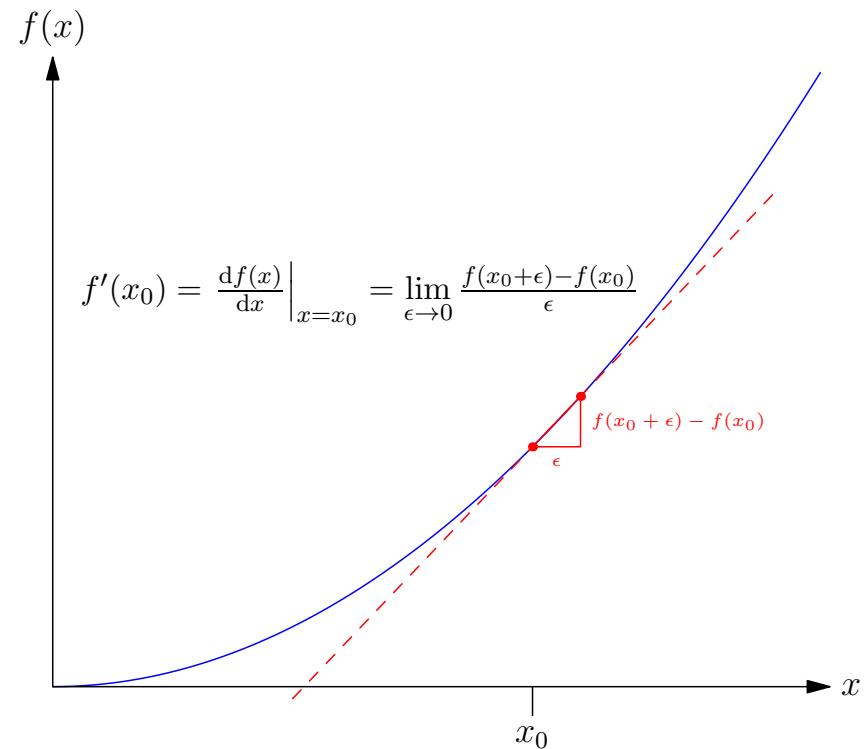
Differential Calculus



Differentiation, product and chain rules, vectors and matrices

Outline

1. **Why Calculus?**
2. Differentiation
3. Vector and Matrix Calculus



Why Calculus?

- Calculus is a fundamental tool of mathematical analysis
- In machine learning differentiation is fundamental tool in optimisation
- Integration is an essential tool in taking expectations over continuous distributions
- Both differentiation and integration crop up elsewhere

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Back to Basics

- You have all done A-level maths so should be familiar with the rules of calculus
- But, it is easy to forget the rules and sometimes we use quite sophisticated tricks
- Although the sophisticated tricks really speed up calculations, it pays to be able to understand where these tricks come from

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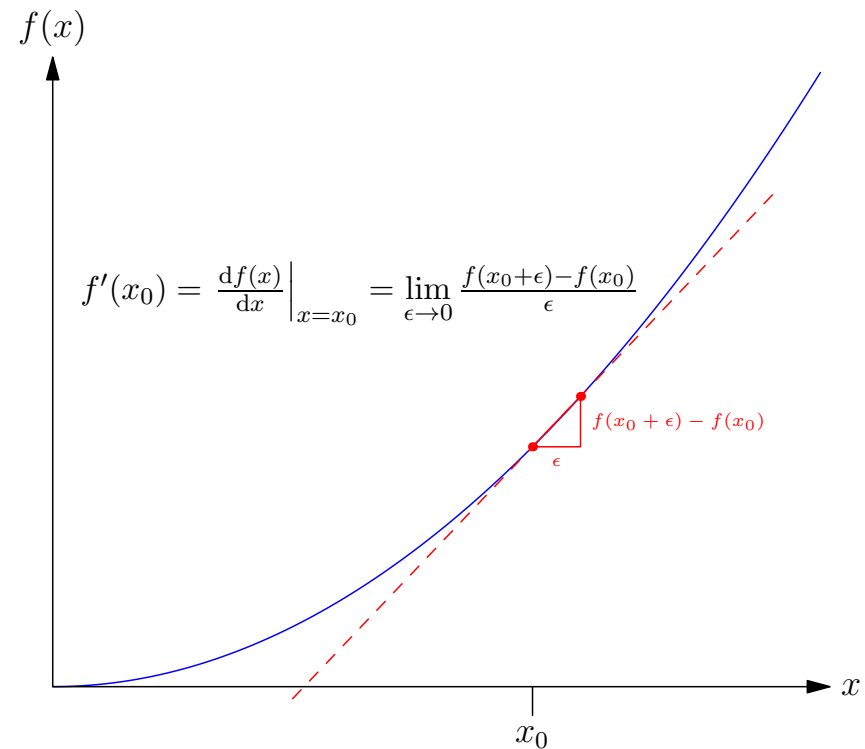
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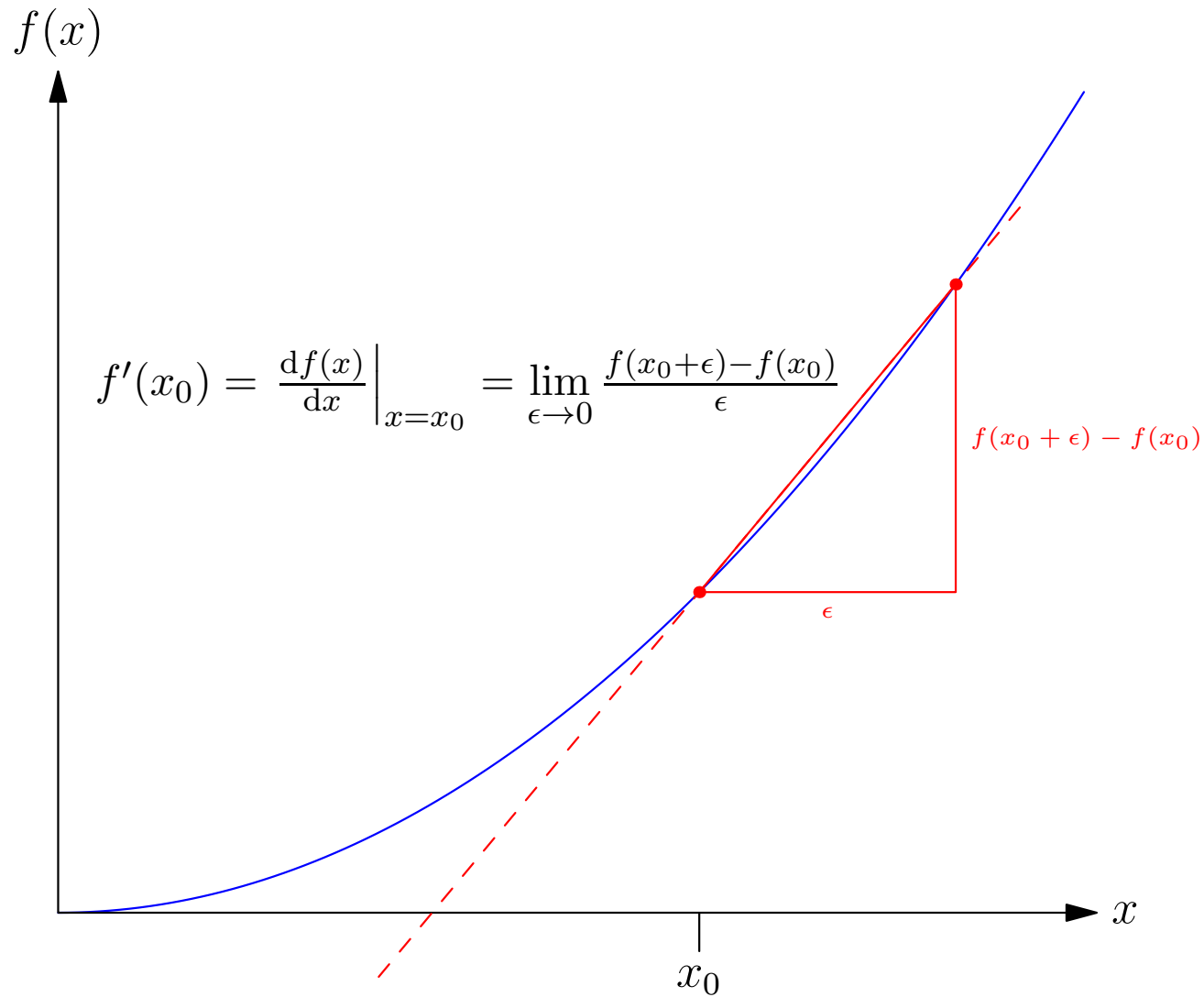
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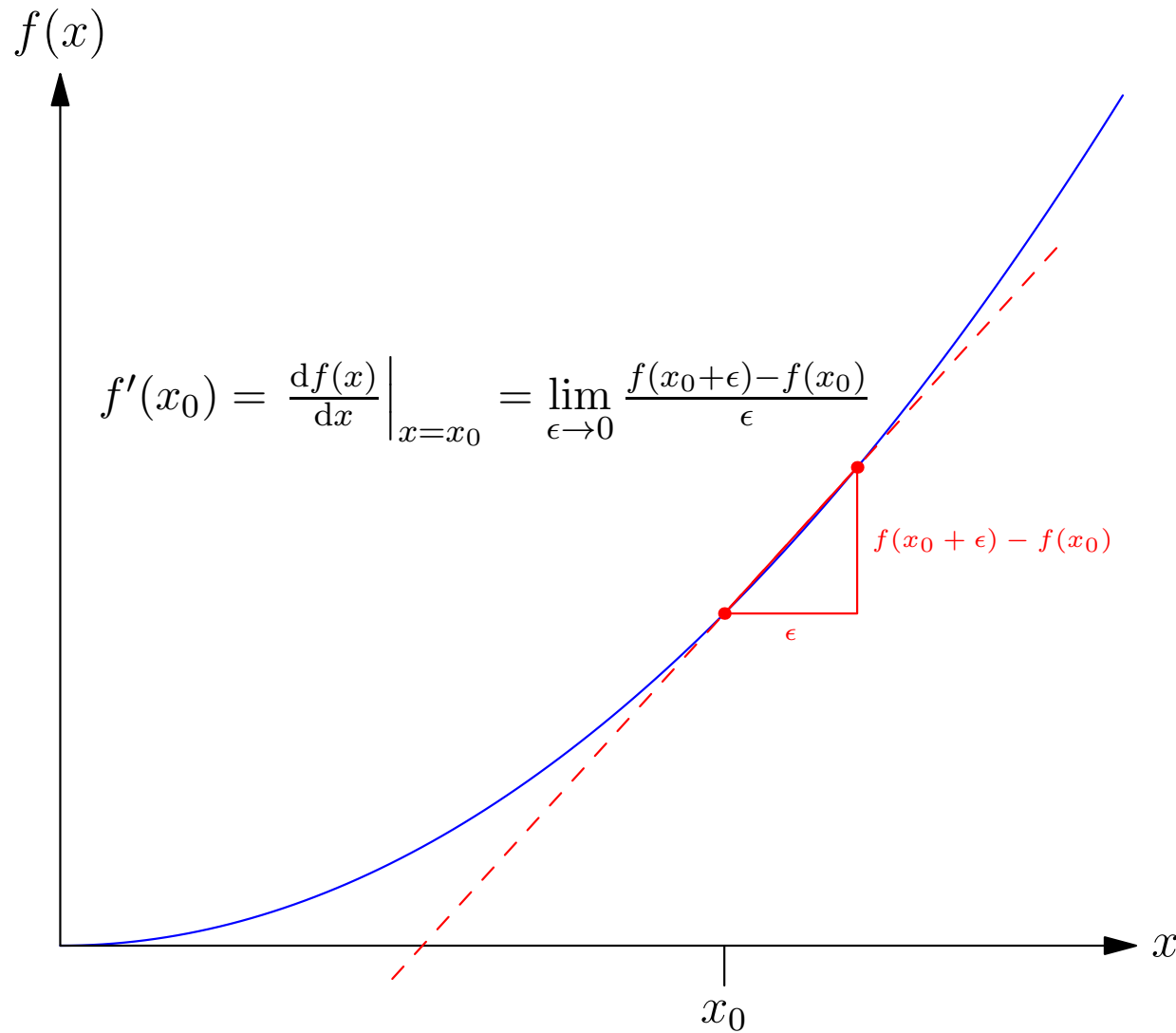
1. Why Calculus?
2. **Differentiation**
3. Vector and Matrix Calculus



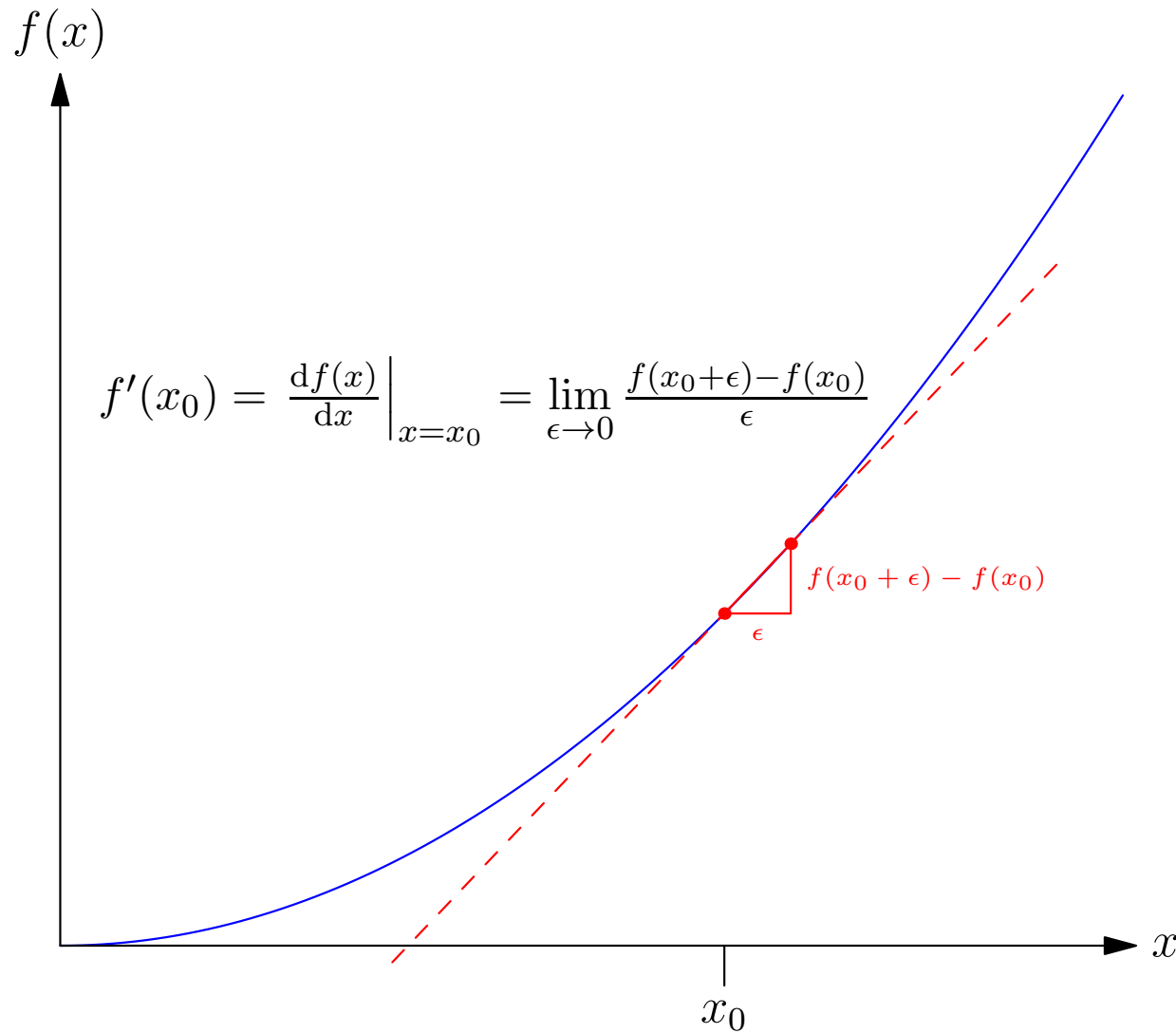
Differentiation



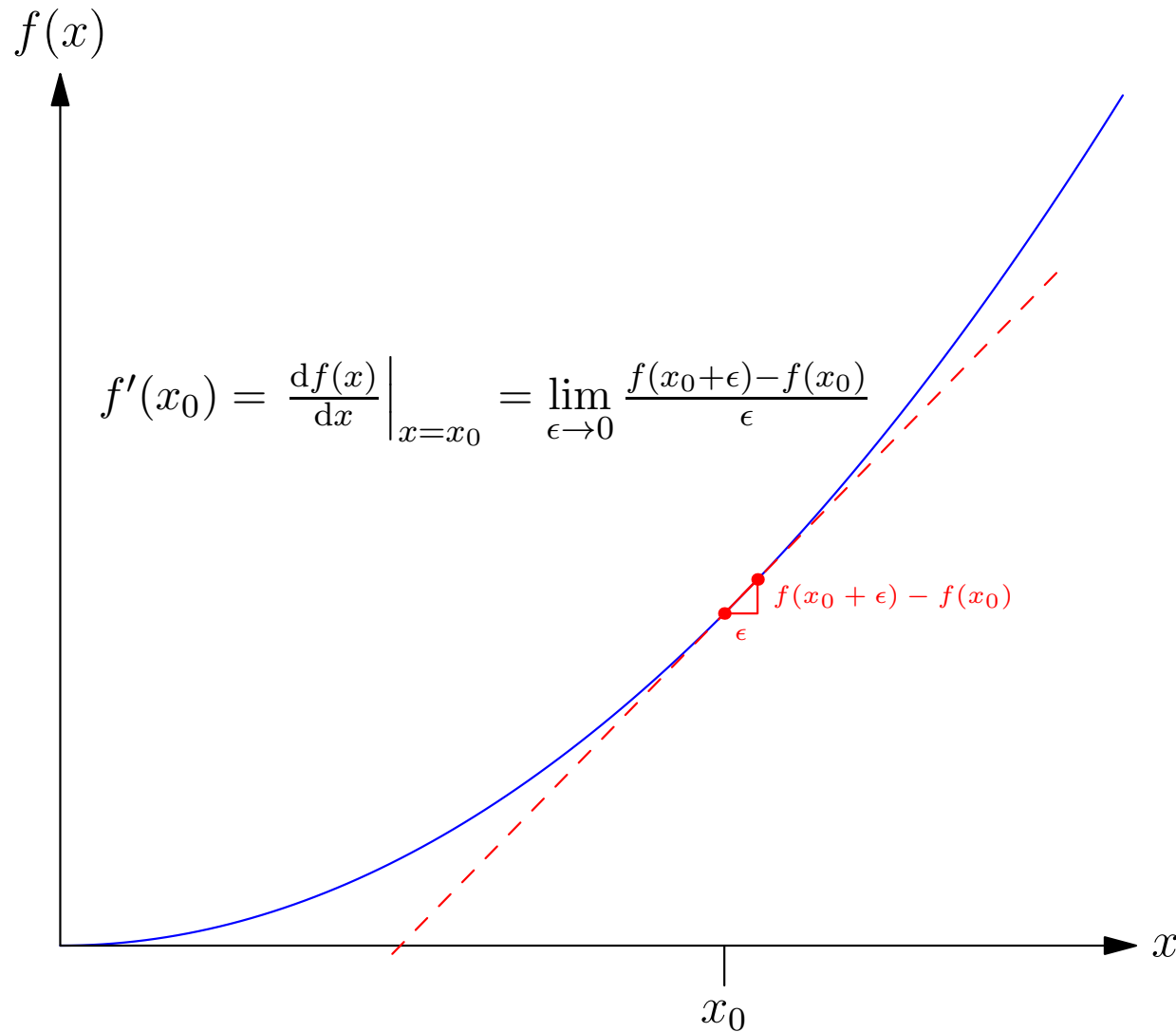
Differentiation



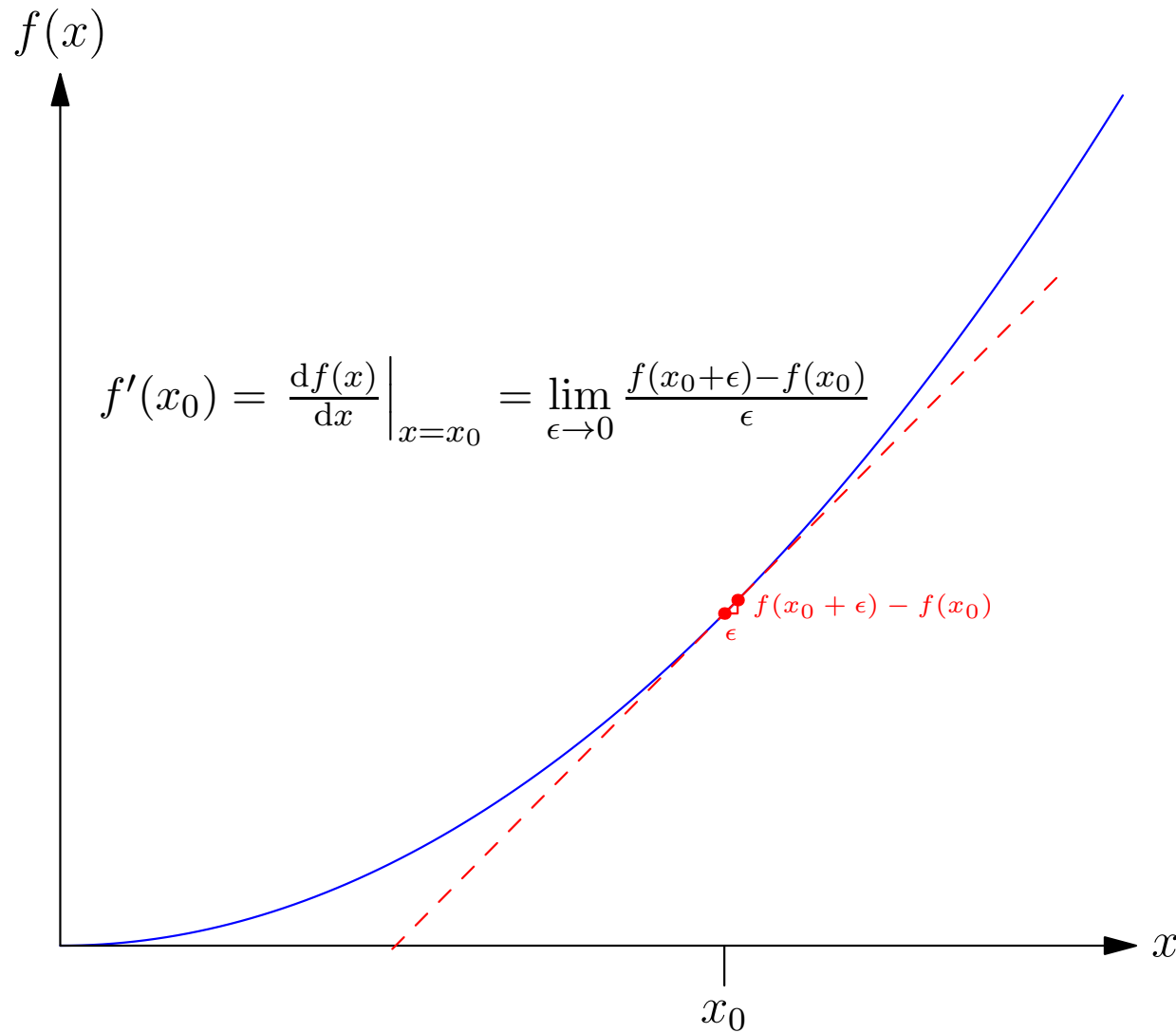
Differentiation



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Differentiation



Polynomials

- $f(x) = x^2$

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Linearity of derivatives

- Consider the derivative of $af(x) + bg(x)$

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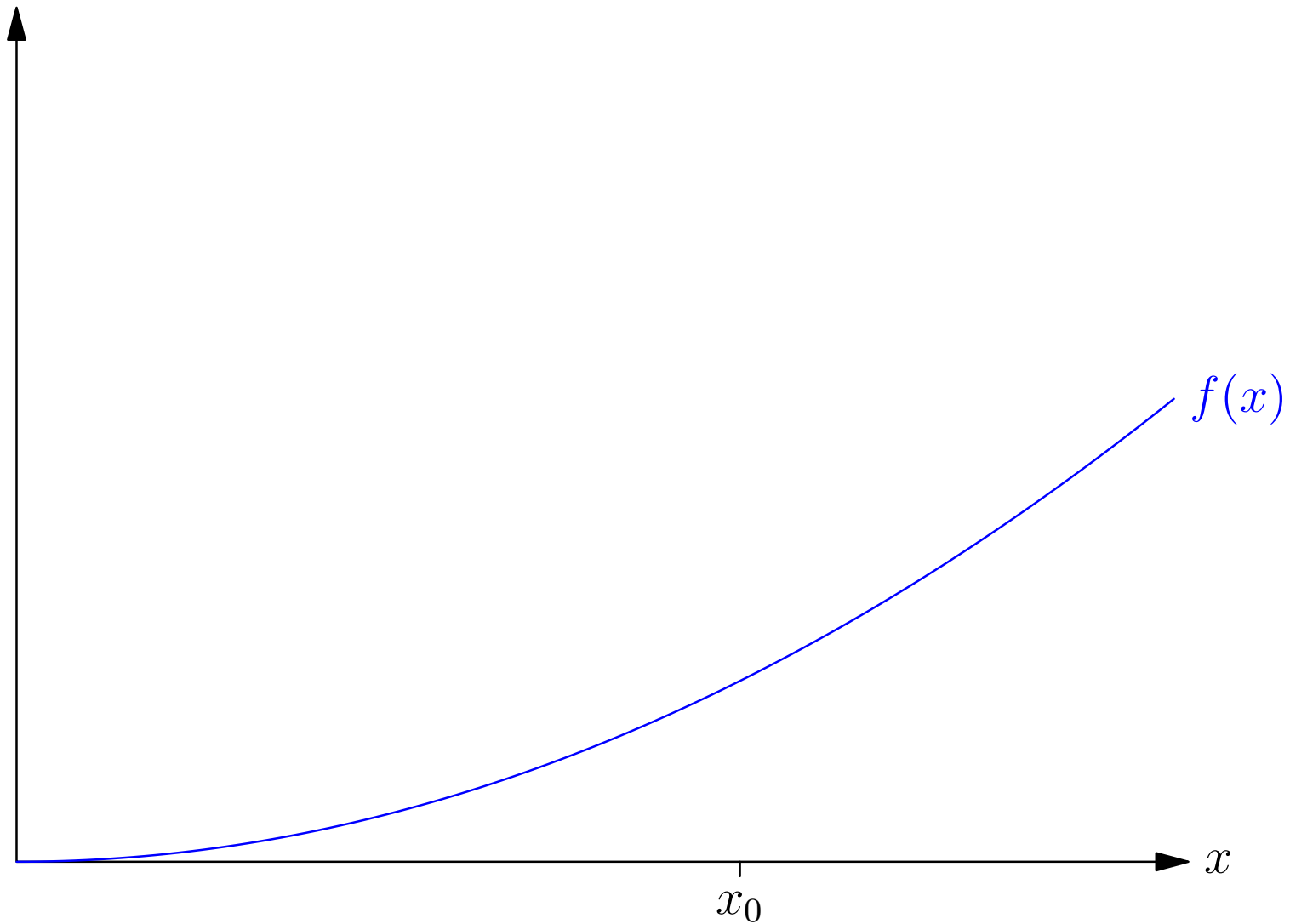
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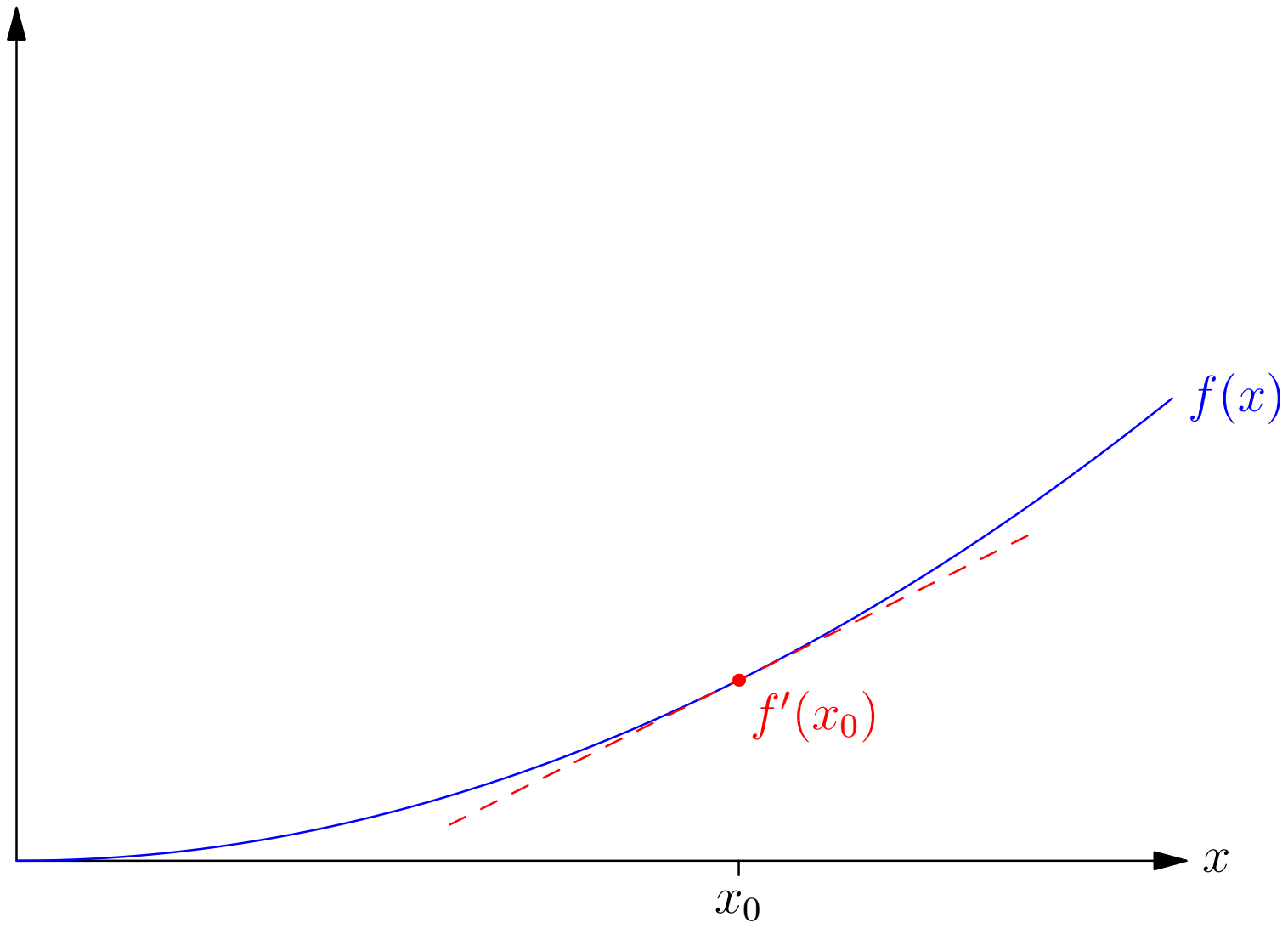
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- Differentiation is a linear operation!**

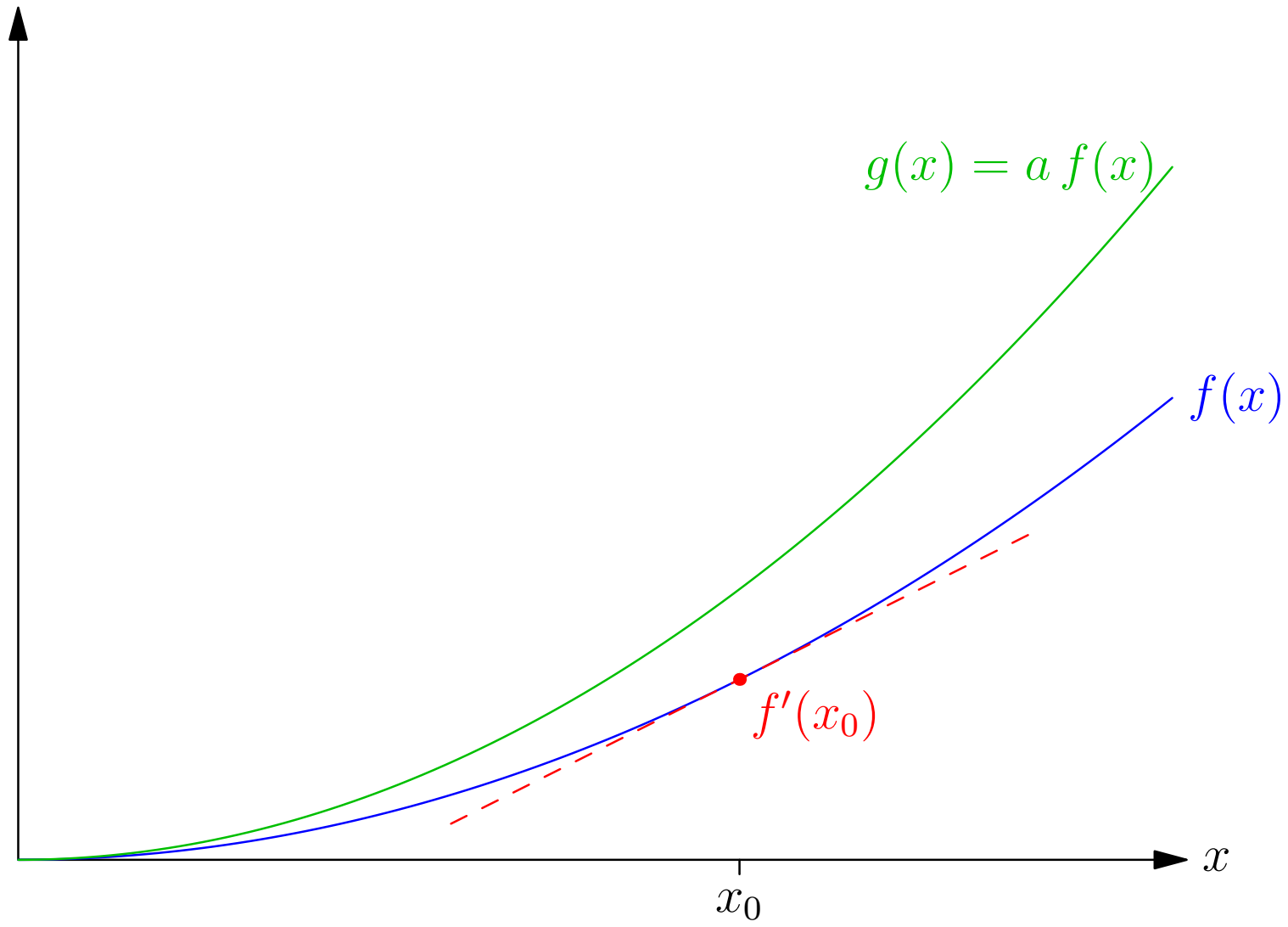
Linearity in Pictures



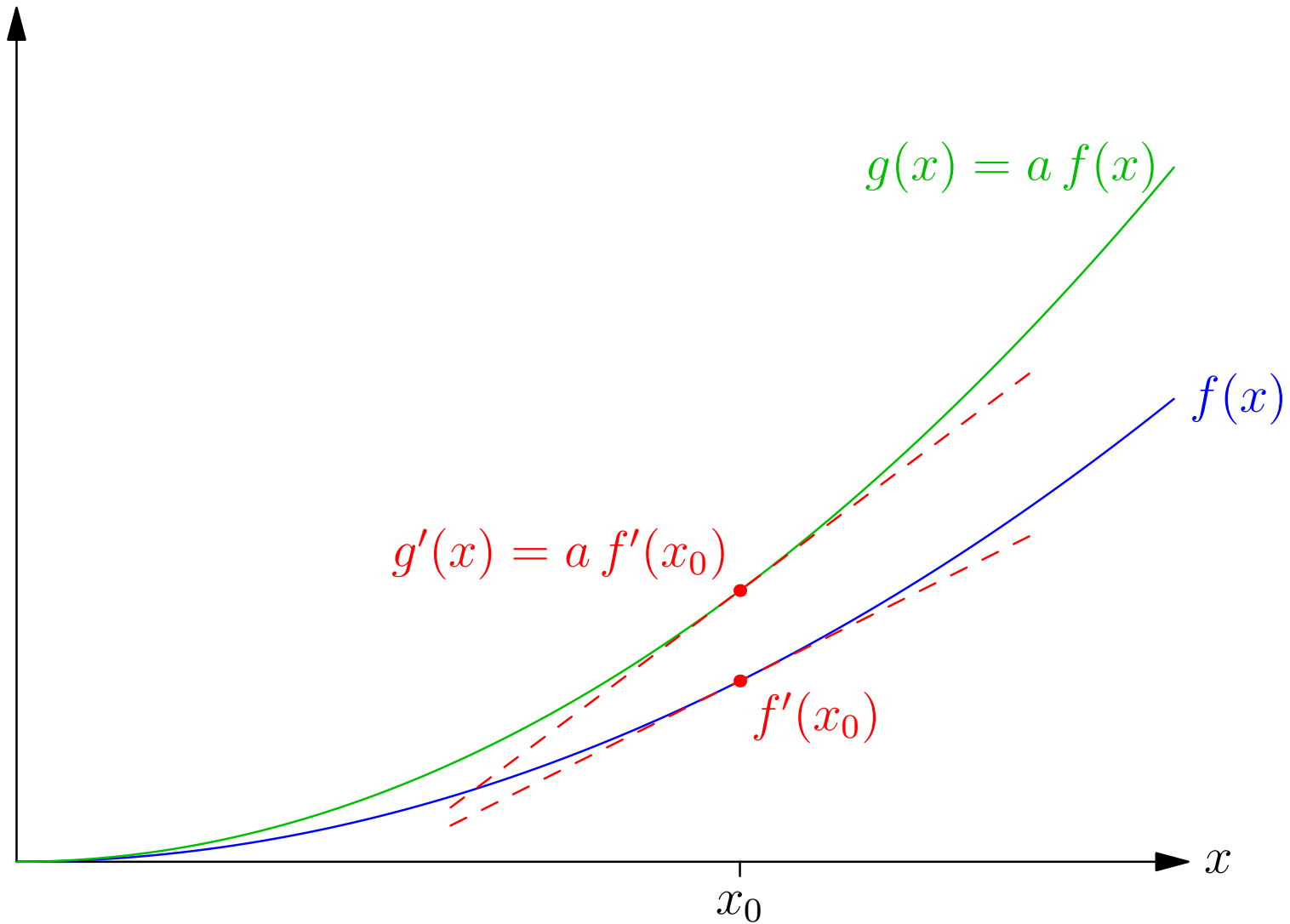
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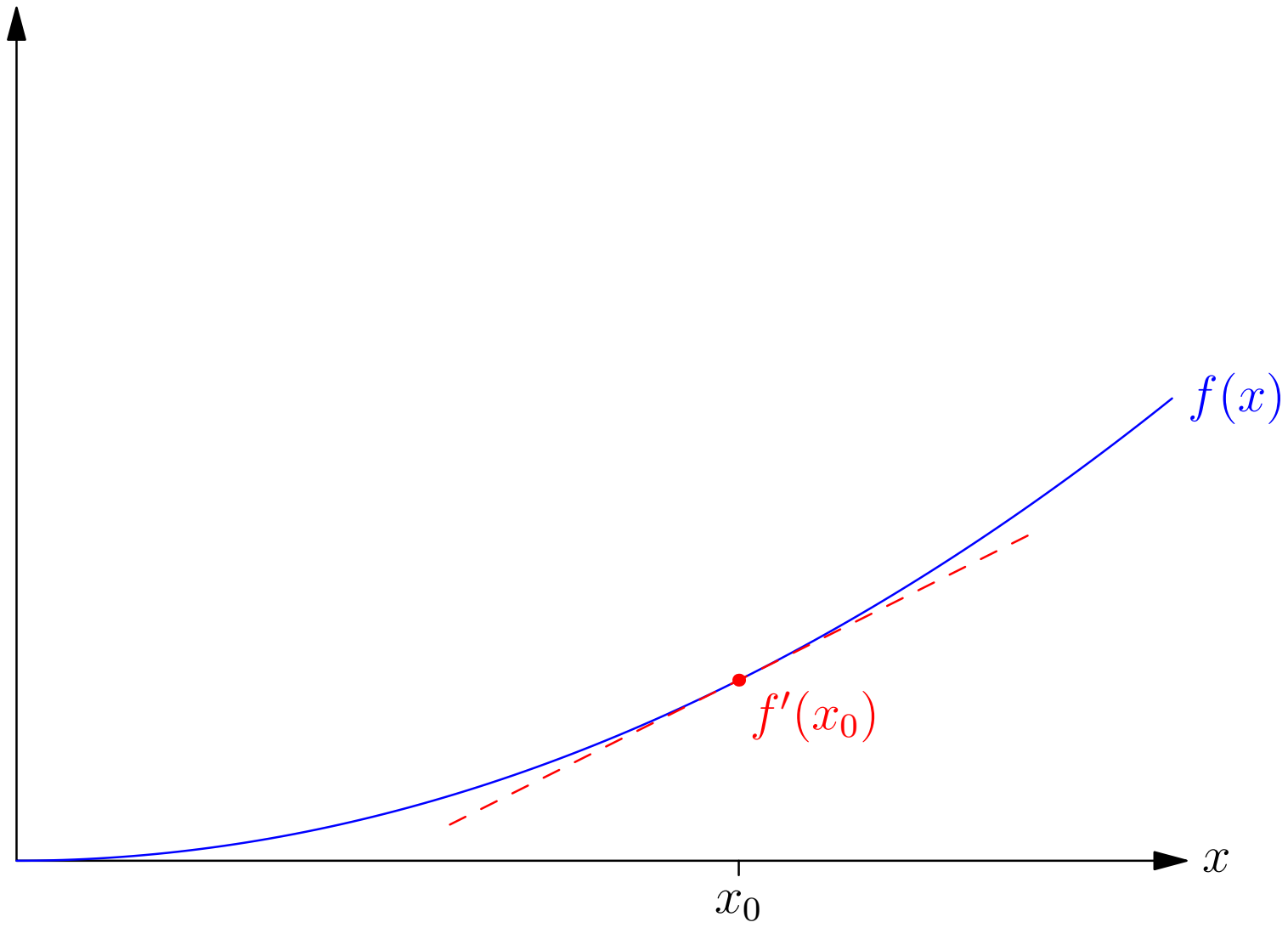
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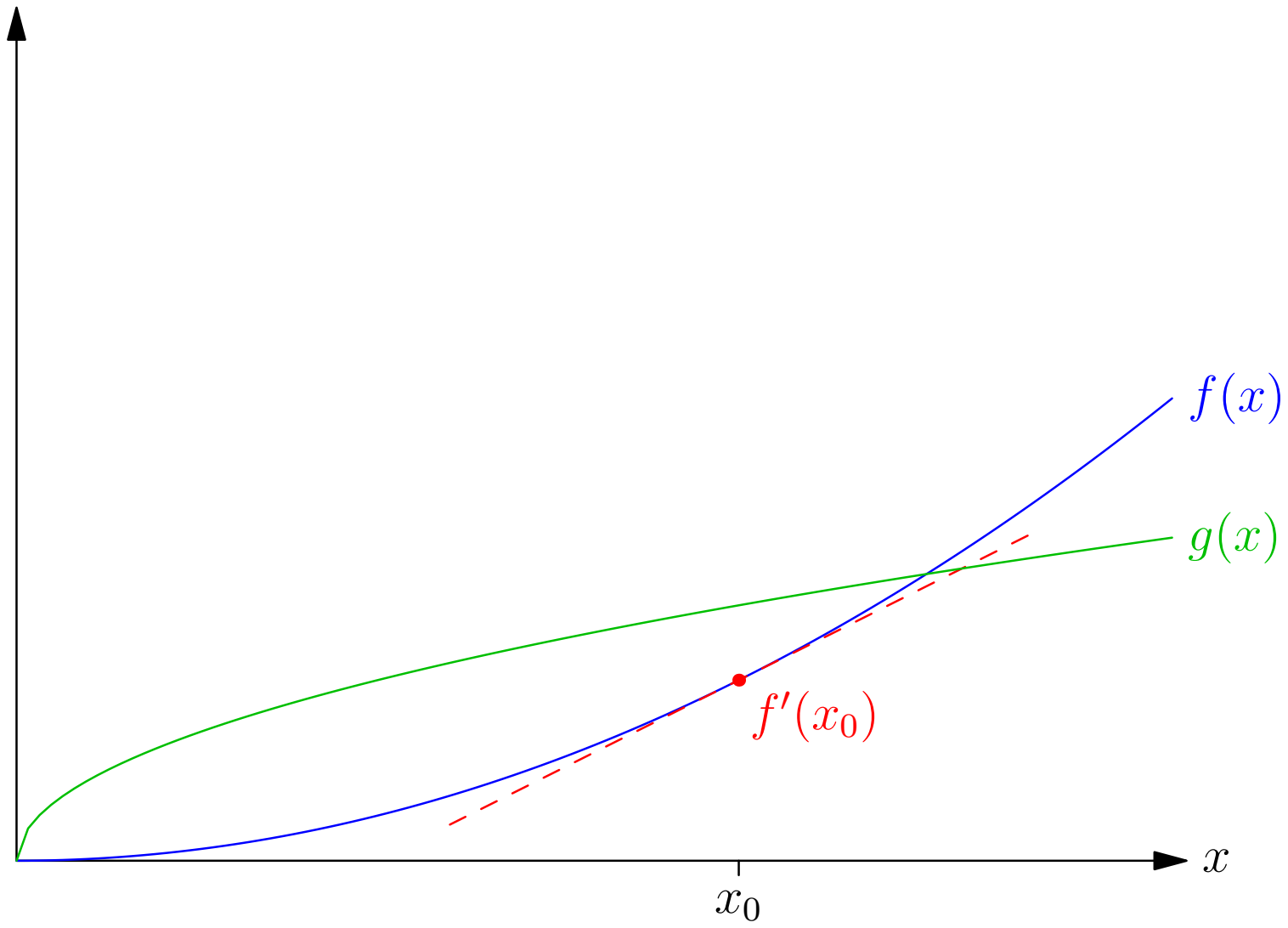
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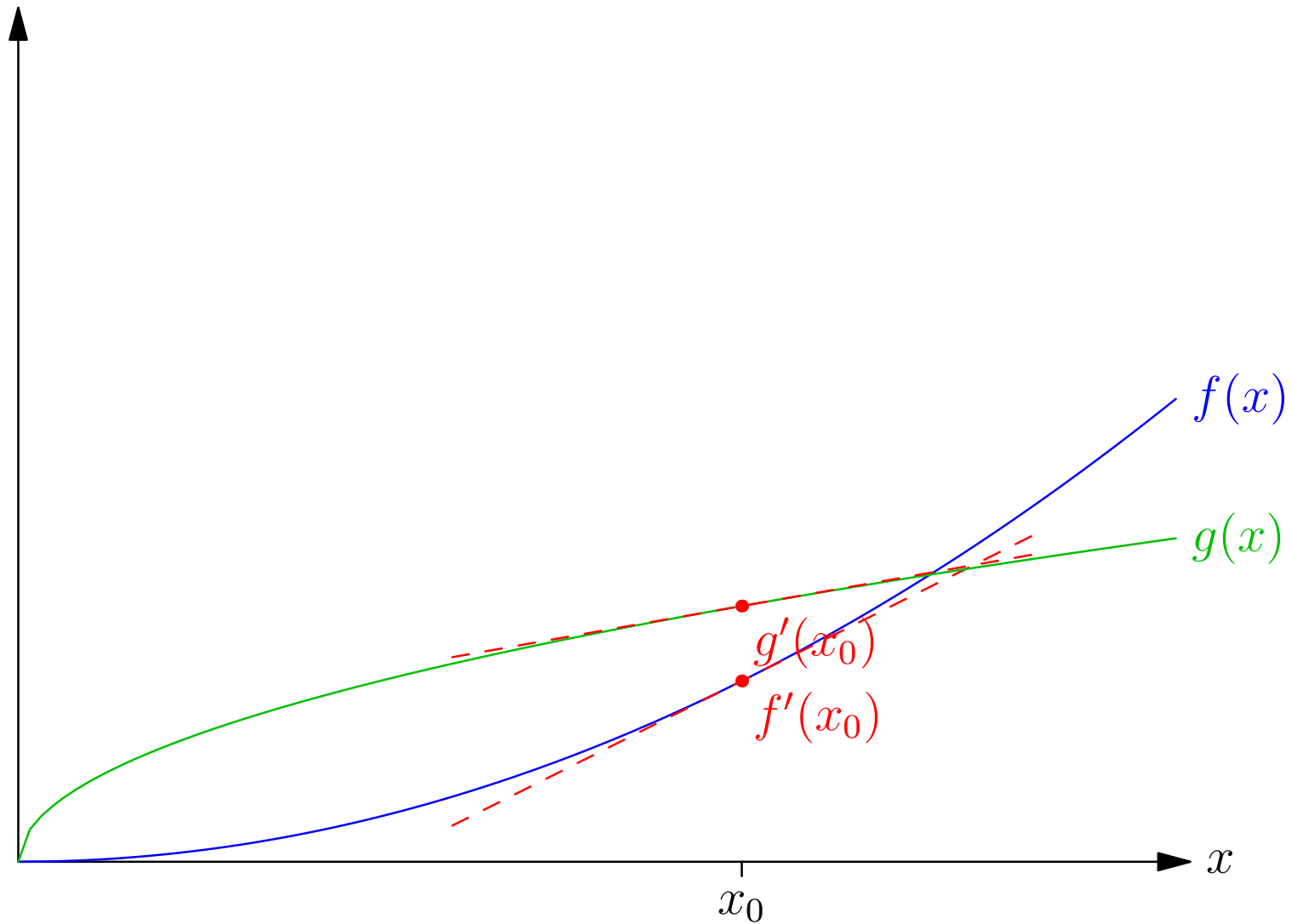
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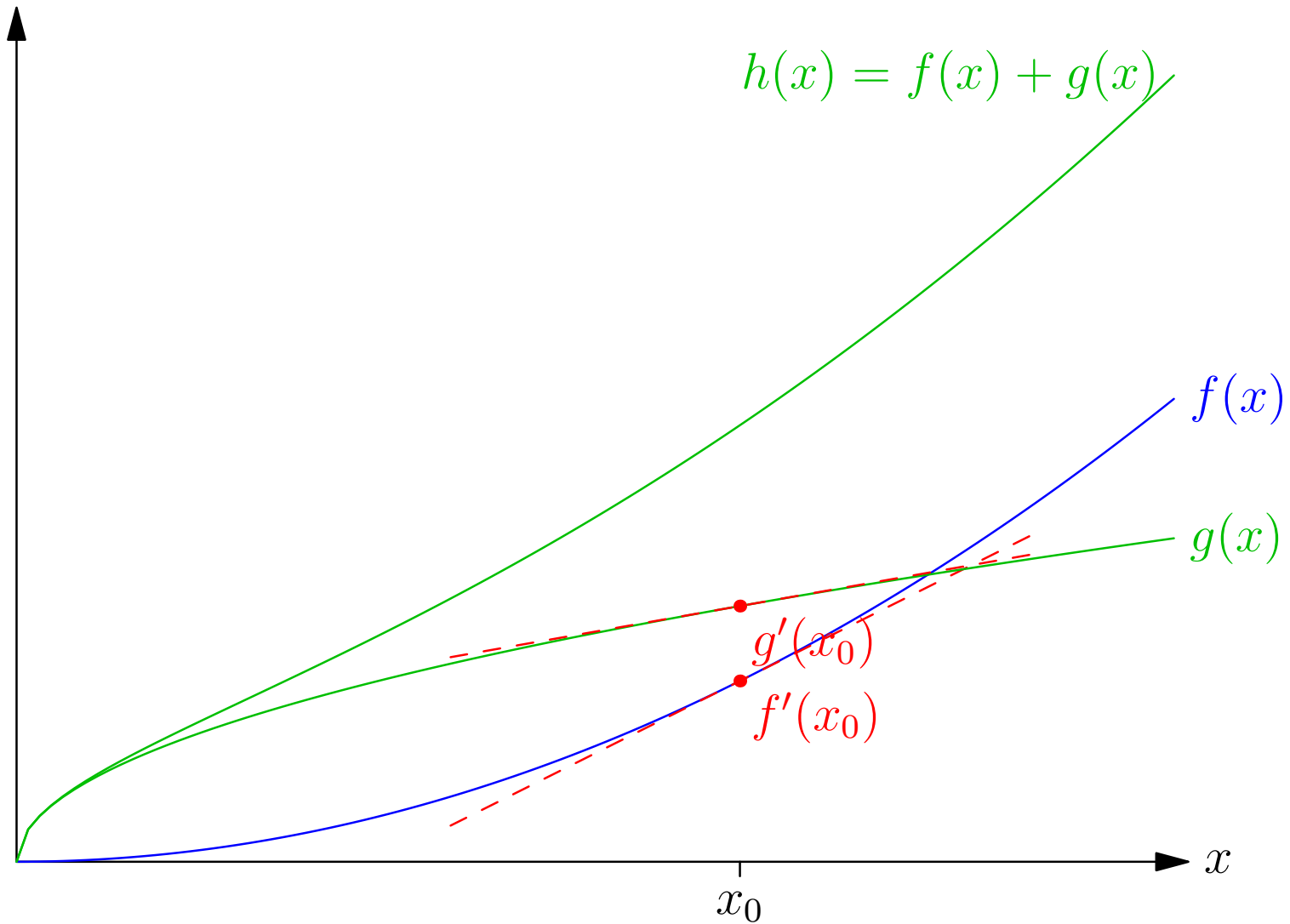
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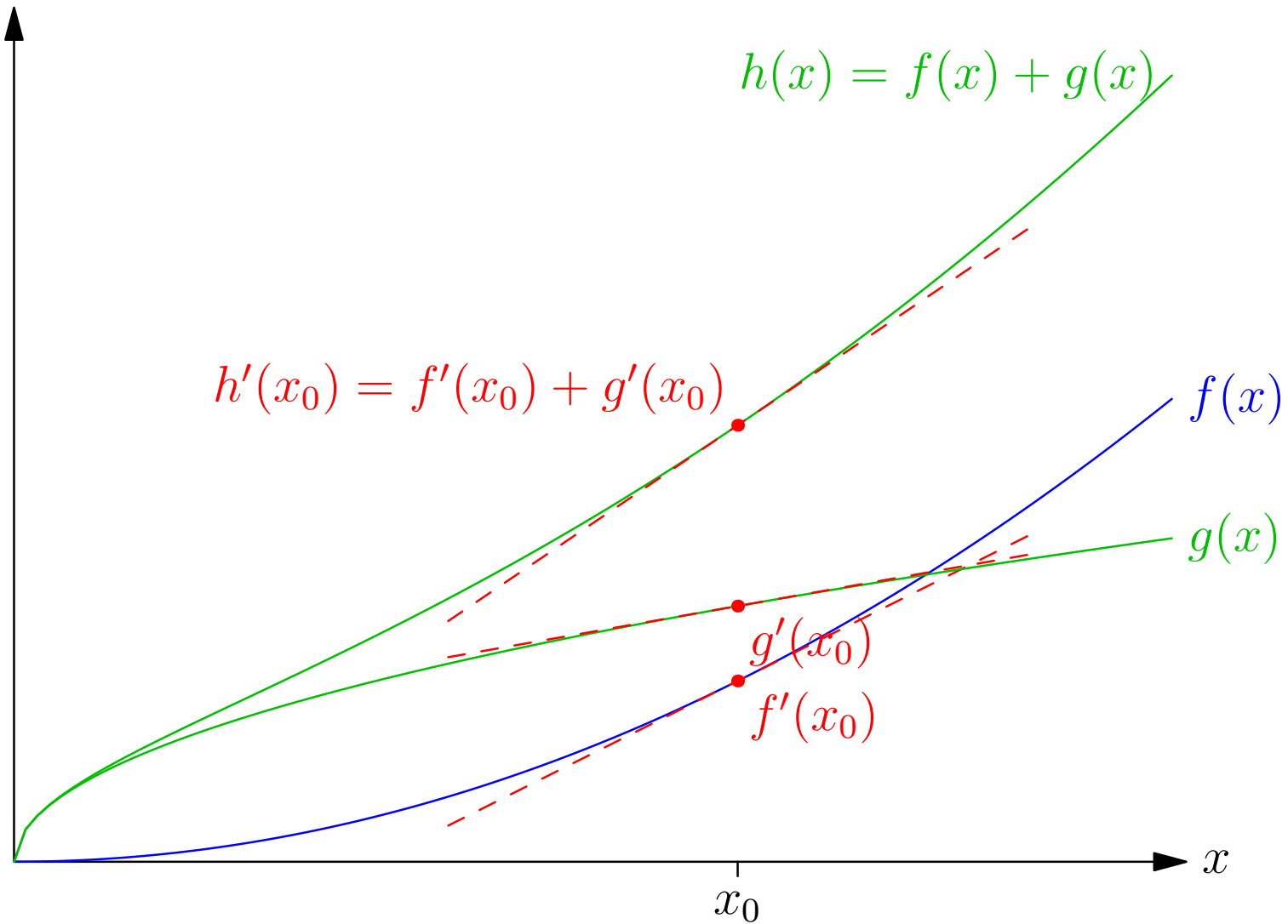
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- This is the famous **chain rule**. Together with the product rule it means you can differentiate almost everything

More on chain rules

- We can also write the chain rule as

$$\frac{df(g(x))}{dx} = \frac{df(g)}{dg} \frac{dg(x)}{dx}$$

- Sometimes this is neater or easier to remember

$$\frac{de^{\cos(x^2)}}{dx} = \frac{de^{\cos(x^2)}}{d\cos(x^2)} \frac{d\cos(x^2)}{dx^2} \frac{dx^2}{dx}$$

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Inverse functions

- Suppose $g(y) = f^{-1}(y)$ is the inverse of $f(x)$ in the sense that $g(f(x)) = f^{-1}(f(x)) = x$
- Using the chain rule

$$\frac{dg(f(x))}{dx} = f'(x)g'(f(x))$$

- So $g'(f(x)) = 1/f'(x)$
- Writing $y = f(x)$ so that $x = f^{-1}(y) = g(y)$ we find $g'(y) = 1/f'(g(y))$ that is

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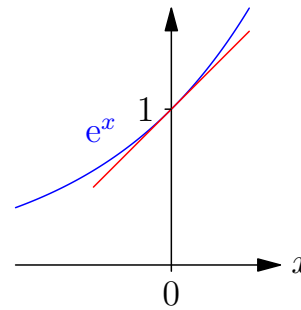
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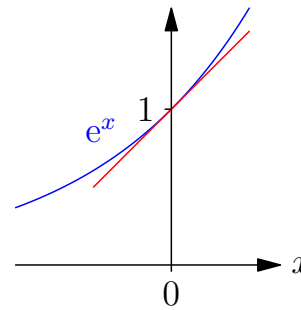
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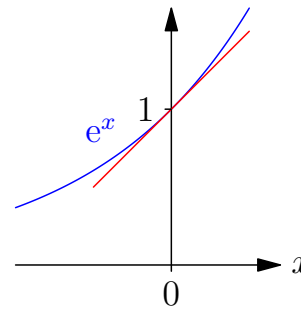


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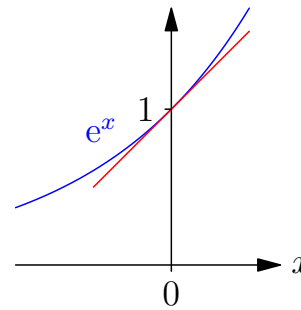


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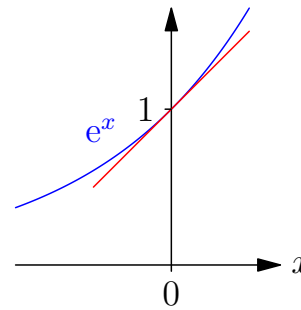
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Functions of Exponentials

- What about $f(x) = e^{cx}$

$$\frac{de^{cx}}{dx} = \frac{de^{cx}}{dcx} \frac{dcx}{dx}$$

- More generally using the chain rule

$$\frac{de^{g(x)}}{dx} = g'(x)e^{g(x)}$$

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Natural Logarithms

- The natural logarithm is defined as the inverse of e^x

$$\ln(e^x) = x \qquad e^{\ln(y)} = y$$

- Recall that if $g(y) = f^{-1}(y)$ then $g'(y) = 1/f'(g(y))$
- Consider $g(y) = \ln(y)$ and $f(x) = e^x$ (with $f'(x) = e^x$)

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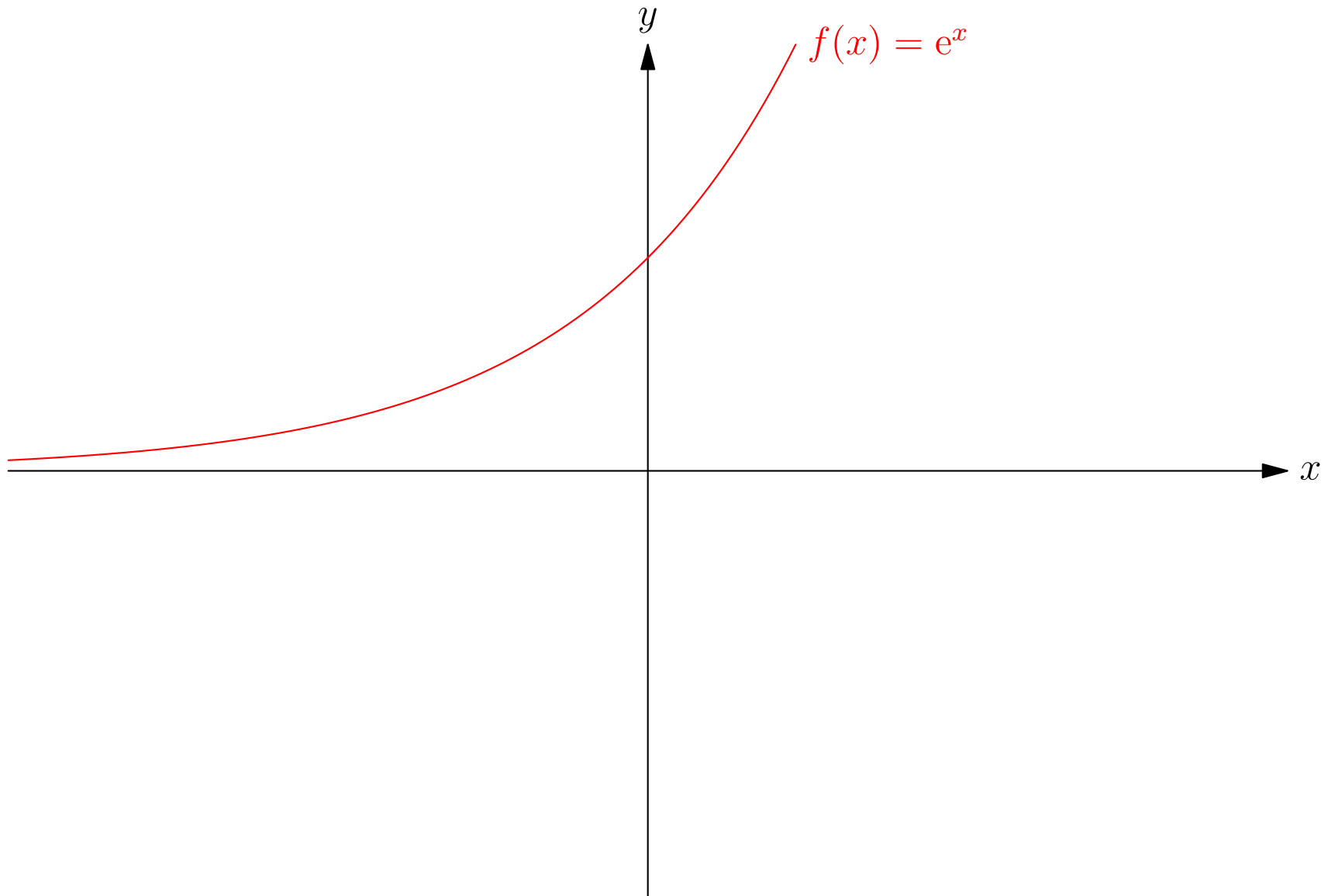
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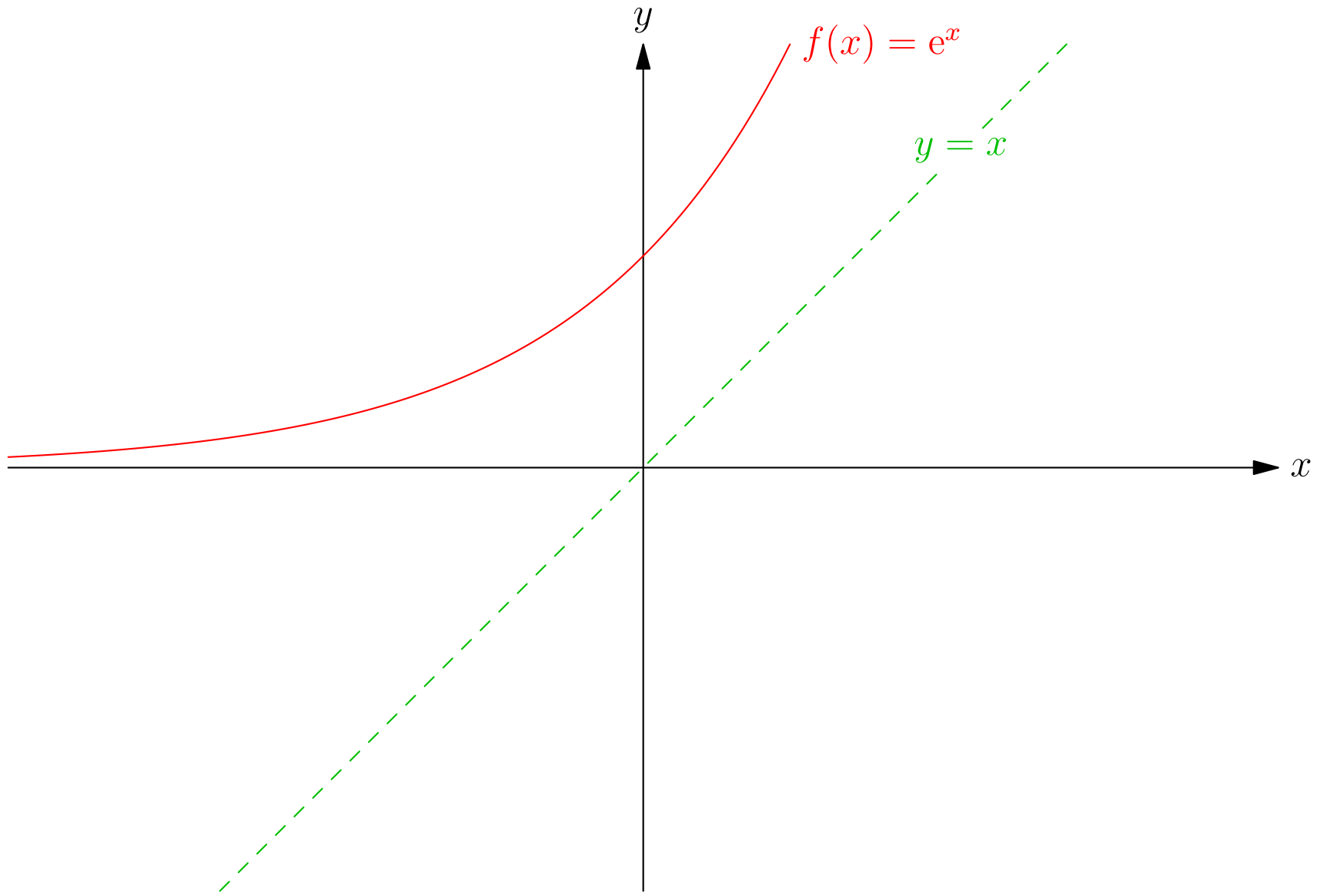
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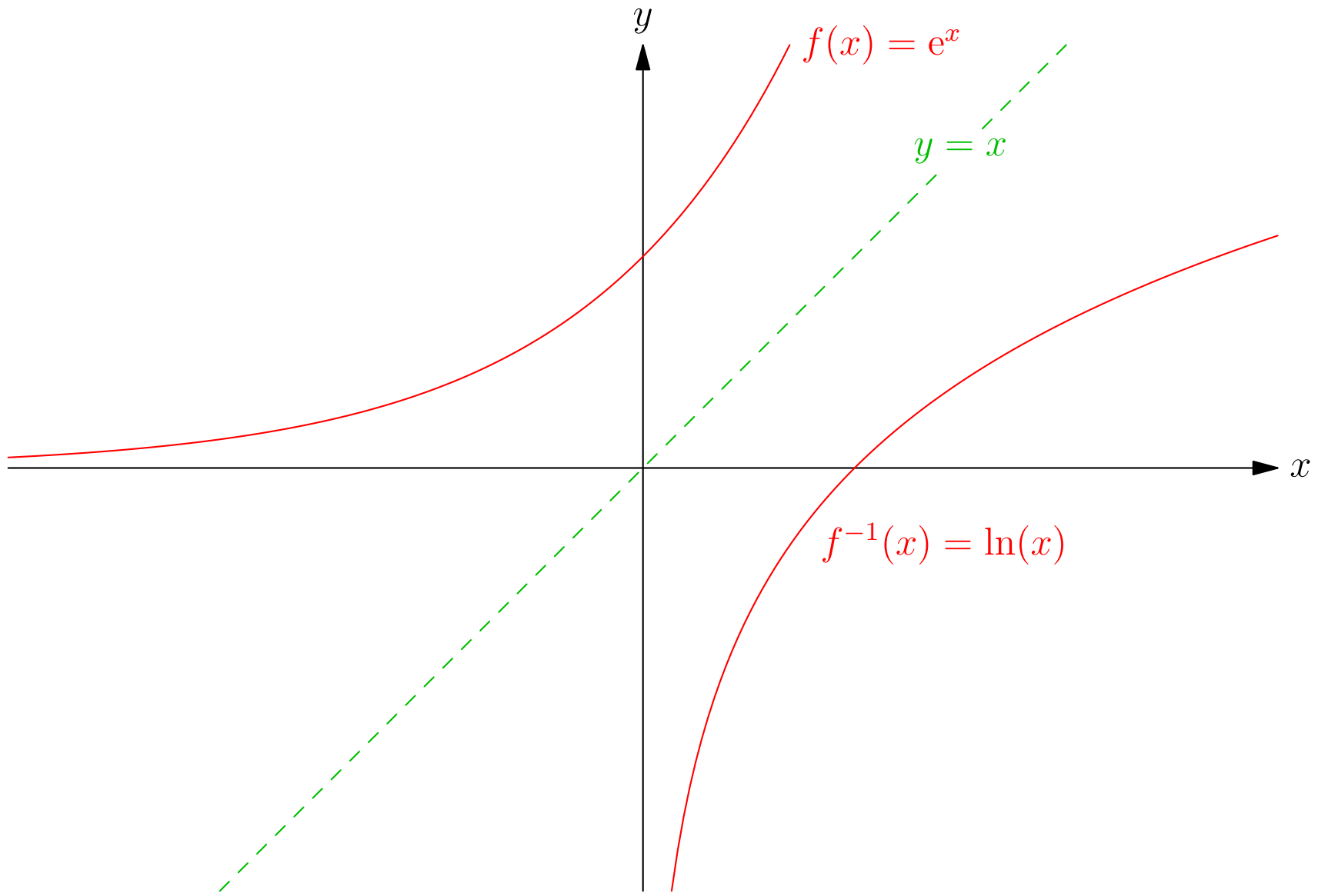
Exponentials and Logarithms



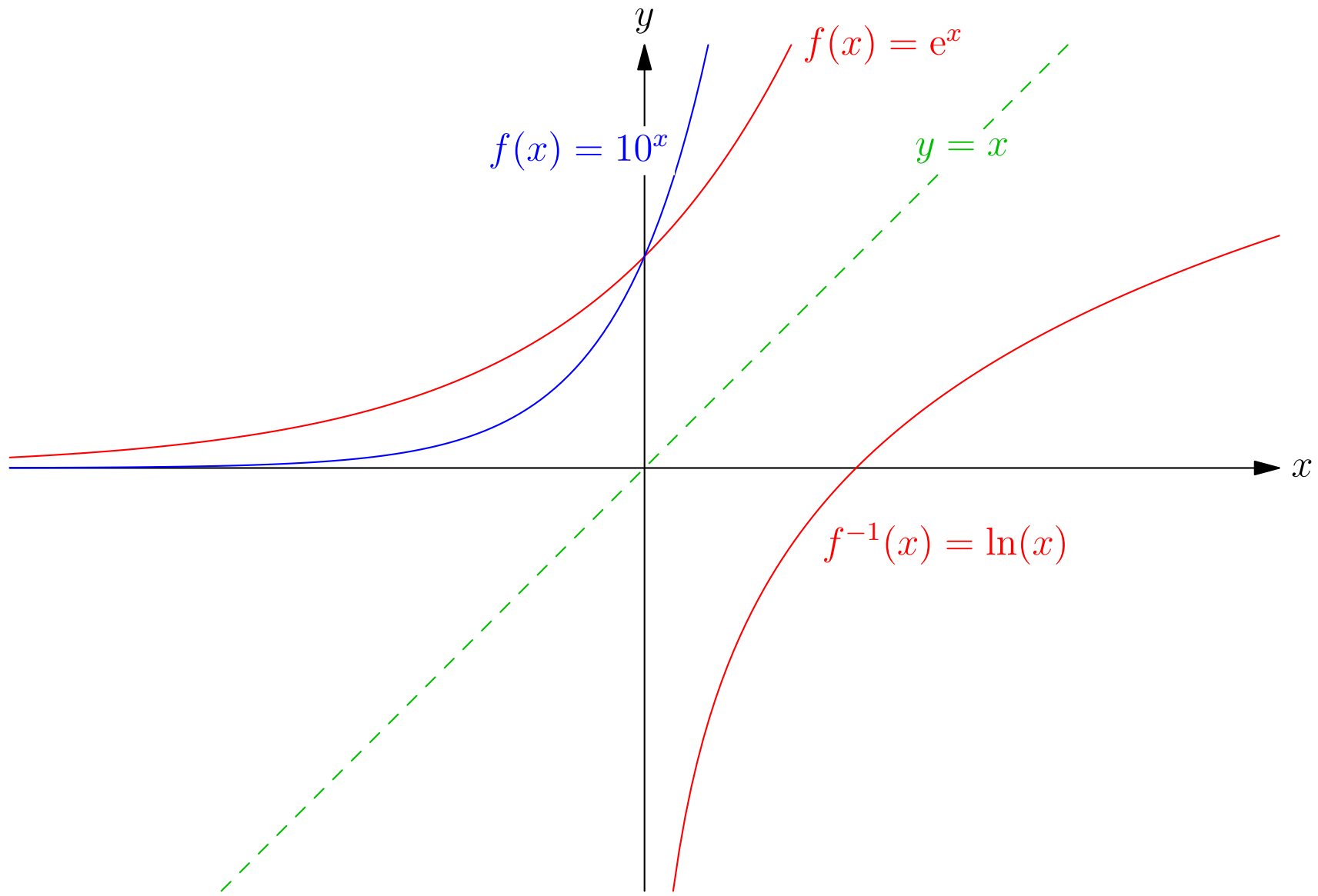
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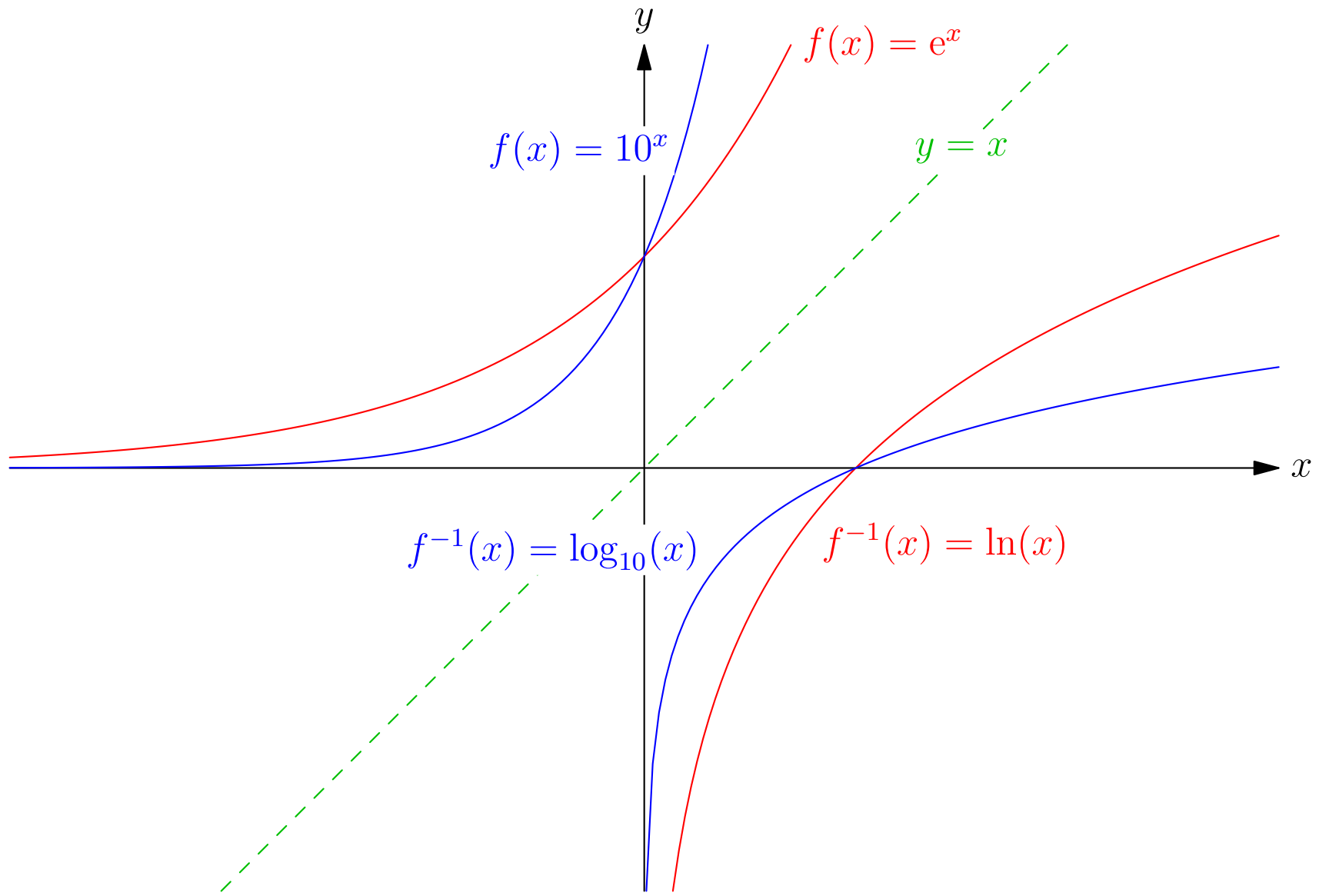
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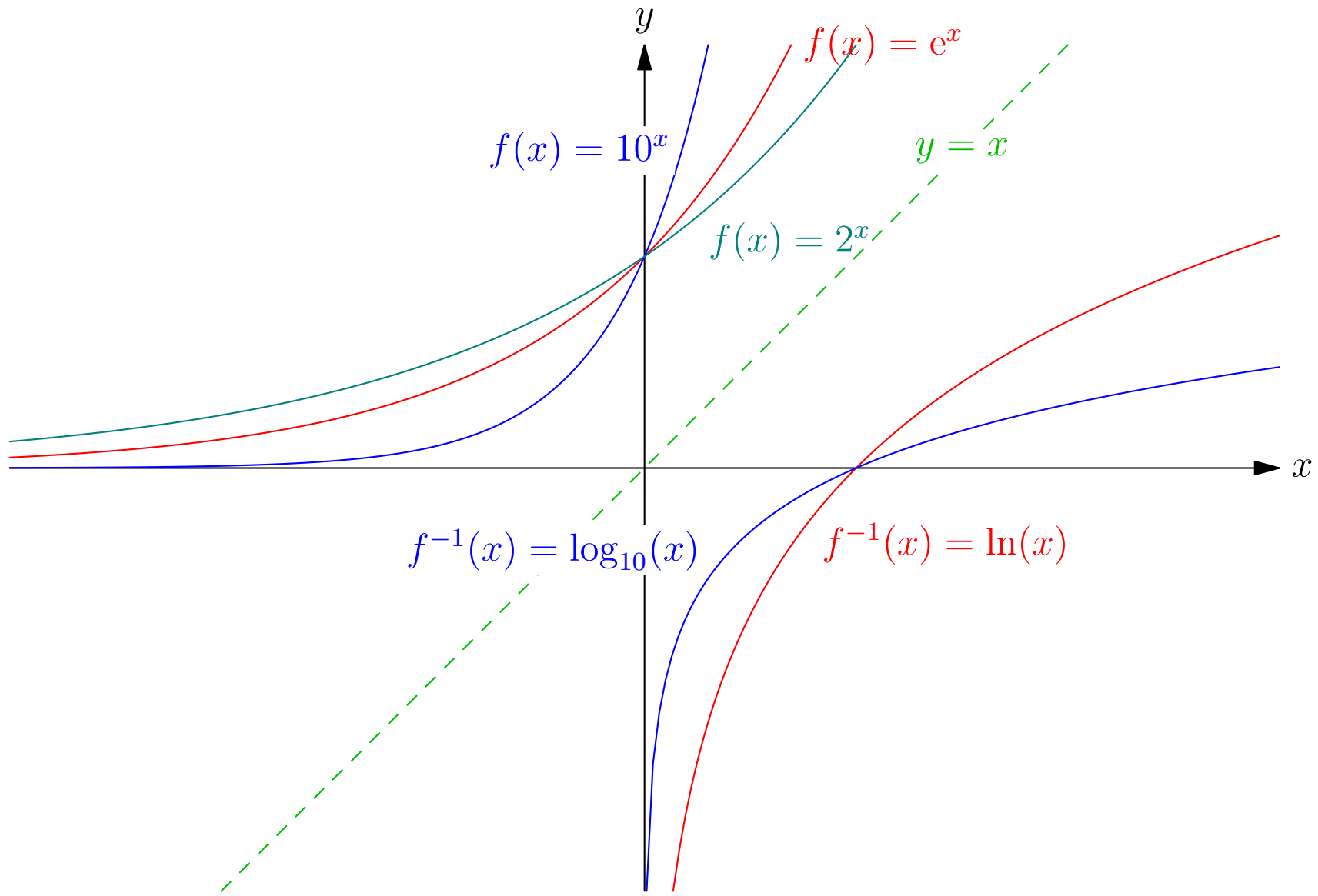
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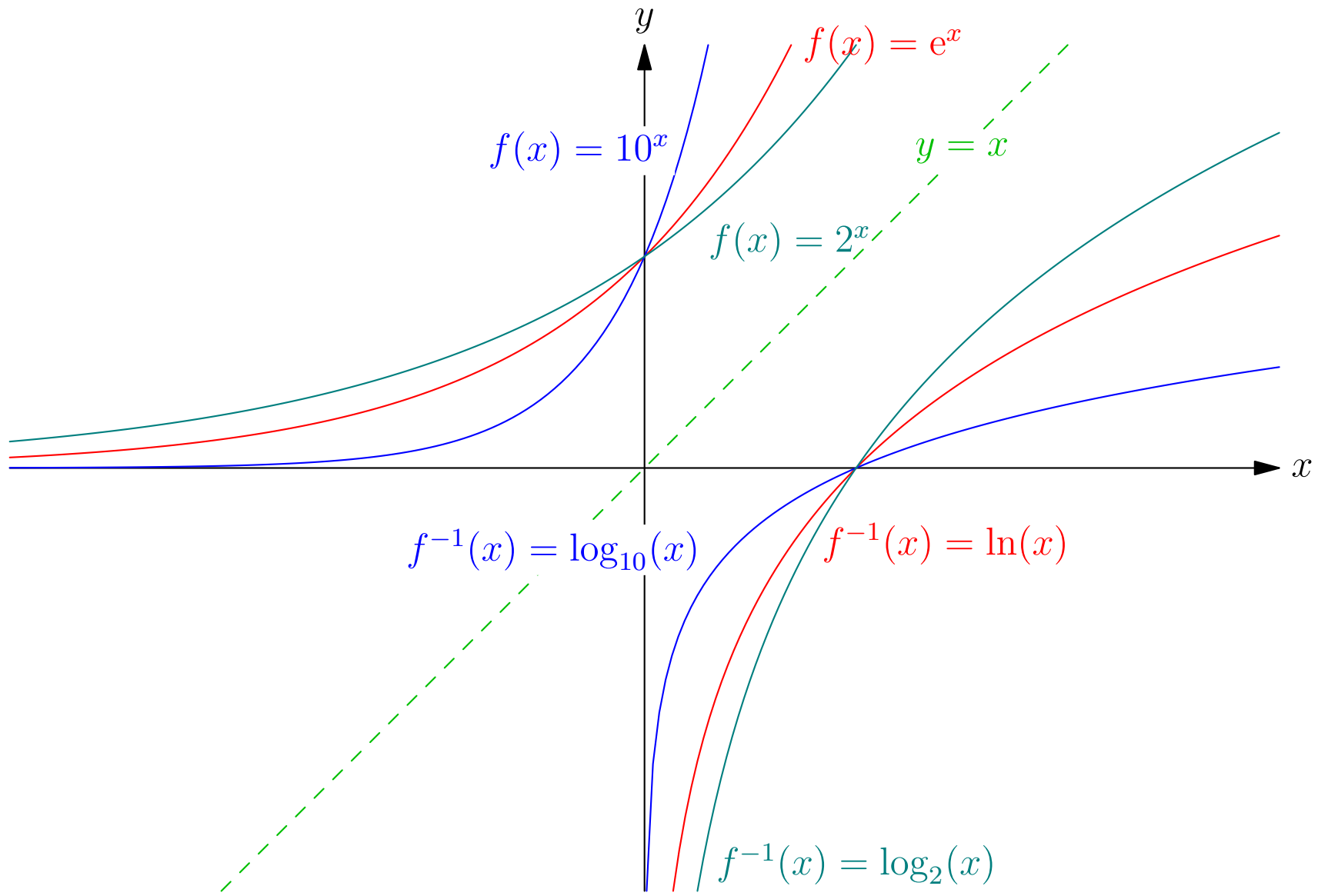
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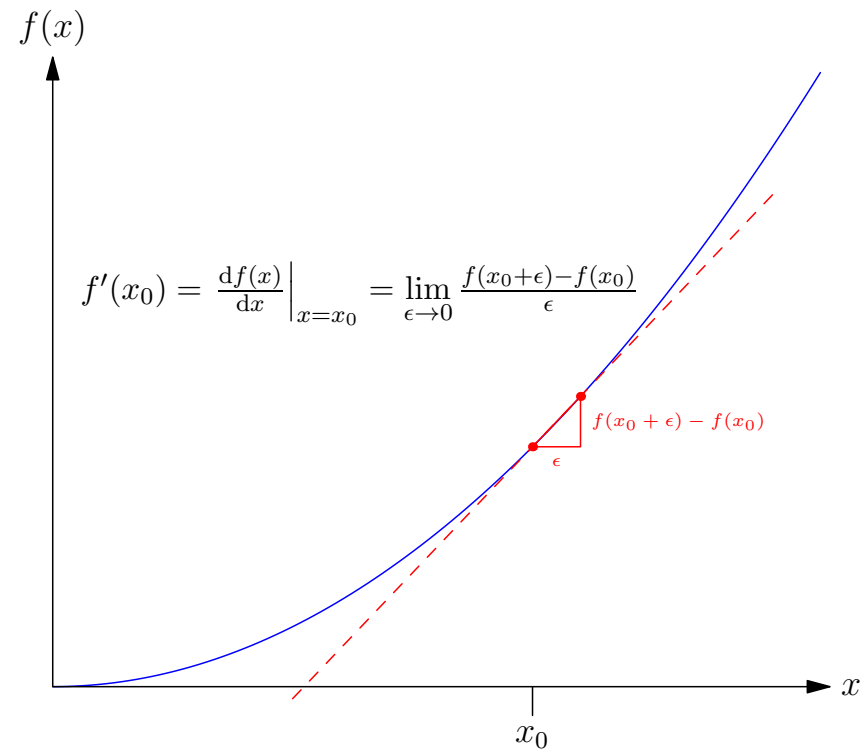


Exponentials and Logarithms



Outline

1. Why Calculus?
2. Differentiation
3. **Vector and Matrix Calculus**



Derivatives in High Dimensions

- When working with functions $f : \mathbb{R}^n \rightarrow \mathbb{R}$ in many dimensions then there will typically be different derivative in different directions
- To compute the derivative in a direction $\mathbf{u} \in \mathbb{R}^n$ (where $\|\mathbf{u}\| = 1$) at a point $\mathbf{x} \in \mathbb{R}^n$ we use

$$\partial_{\mathbf{u}} F(\mathbf{x}) = \lim_{\epsilon \rightarrow 0} \frac{f(\mathbf{x} + \epsilon \mathbf{u}) - f(\mathbf{x})}{\epsilon}$$

- If $\mathbf{u} = \boldsymbol{\delta}_i = (0, \dots, 0, 1, 0, \dots, 0)$ (i.e. $u_i = 1$) then

$$\frac{\partial f(\mathbf{x})}{\partial x_i} = \lim_{\epsilon \rightarrow 0} \frac{f(\mathbf{x} + \epsilon \boldsymbol{\delta}_i) - f(\mathbf{x})}{\epsilon}$$

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Taylor

- If we expand $f(\mathbf{x} + \epsilon \mathbf{u})$ to first order in ϵ

$$f(\mathbf{x} + \epsilon \mathbf{u}) = f(\mathbf{x}) + \epsilon \mathbf{u}^\top \mathbf{g}(\mathbf{x}) + O(\epsilon^2)$$

then $g_i(\mathbf{x}) = \frac{\partial f(\mathbf{x})}{\partial x_i}$

- Recall we defined the vector of first order derivatives of $f(\mathbf{x})$ to be the gradient

$$\nabla f(\mathbf{x}) = \begin{pmatrix} \frac{\partial f(\mathbf{x})}{\partial x_1} \\ \frac{\partial f(\mathbf{x})}{\partial x_2} \\ \vdots \\ \frac{\partial f(\mathbf{x})}{\partial x_n} \end{pmatrix}$$

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This is the start of the high-dimensional Taylor expansion

Computing Gradients 1

- We can compute the gradient by writing out $f(x)$ componentwise and performing the partial derivative with respect to x_i

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- It is tedious to compute these things component-wise, but when you need to understand what is going on then go back to the basics

Computing Gradients 2

- A slicker way is just to expand $f(\mathbf{x} + \epsilon \mathbf{u})$
- Consider $f(\mathbf{x}) = \mathbf{x}^\top \mathbf{M} \mathbf{x} + \mathbf{a}^\top \mathbf{x}$

$$f(\mathbf{x} + \epsilon \mathbf{u}) = (\mathbf{x} + \epsilon \mathbf{u})^\top \mathbf{M} (\mathbf{x} + \epsilon \mathbf{u}) + \mathbf{a}^\top (\mathbf{x} + \epsilon \mathbf{u})$$

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- But $f(\mathbf{x} + \epsilon \mathbf{u}) = f(\mathbf{x}) + \epsilon \mathbf{u}^\top \nabla f(\mathbf{x}) + O(\epsilon^2)$ so

$$\nabla f(\mathbf{x}) = \mathbf{M} \mathbf{x} + \mathbf{M}^\top \mathbf{x} + \mathbf{a}$$

Differentiating Matrices

- Often we have loss functions with respect to a matrix $\mathbf{W}$, e.g.

$$L(\mathbf{W}) = (\mathbf{a}^\top \mathbf{W} \mathbf{b} - c)^2$$

- We might want to find the minimum with respect to $\mathbf{W}$
- This occurs at a point $\mathbf{W}^*$ where $L(\mathbf{W})$ does not increase as we change $\mathbf{W}$ in any way
- That is, we seek a $\mathbf{W}^*$ such that, for any matrices $\mathbf{U}$

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Generalised Gradient

- We can generalise the idea of gradient to matrices

$$\frac{\partial L(\mathbf{W})}{\partial \mathbf{W}} = \begin{pmatrix} \frac{\partial L(\mathbf{W})}{\partial W_{11}} & \frac{\partial L(\mathbf{W})}{\partial W_{12}} & \cdots & \frac{\partial L(\mathbf{W})}{\partial W_{1m}} \\ \frac{\partial L(\mathbf{W})}{\partial W_{21}} & \frac{\partial L(\mathbf{W})}{\partial W_{22}} & \cdots & \frac{\partial L(\mathbf{W})}{\partial W_{2m}} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial L(\mathbf{W})}{\partial W_{n1}} & \frac{\partial L(\mathbf{W})}{\partial W_{n2}} & \cdots & \frac{\partial L(\mathbf{W})}{\partial W_{nm}} \end{pmatrix}$$

- From an identical argument we used for vectors

$$L(\mathbf{W} + \epsilon \mathbf{U}) = L(\mathbf{W}) + \epsilon \text{tr} \mathbf{U}^\top \frac{\partial L(\mathbf{W})}{\partial \mathbf{W}} + O(\epsilon^2)$$

where

$$\text{tr} \mathbf{U}^\top \mathbf{G} = \sum_i [\mathbf{U}^\top \mathbf{G}]_{ii} = \sum_{ij} U_{ji} G_{ji} = \sum_{ij} U_{ij} G_{ij} = \langle \mathbf{U}, \mathbf{G} \rangle$$

Generalised Gradient

- We can generalise the idea of gradient to matrices

$$\frac{\partial L(\mathbf{W})}{\partial \mathbf{W}} = \begin{pmatrix} \frac{\partial L(\mathbf{W})}{\partial W_{11}} & \frac{\partial L(\mathbf{W})}{\partial W_{12}} & \cdots & \frac{\partial L(\mathbf{W})}{\partial W_{1m}} \\ \frac{\partial L(\mathbf{W})}{\partial W_{21}} & \frac{\partial L(\mathbf{W})}{\partial W_{22}} & \cdots & \frac{\partial L(\mathbf{W})}{\partial W_{2m}} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial L(\mathbf{W})}{\partial W_{n1}} & \frac{\partial L(\mathbf{W})}{\partial W_{n2}} & \cdots & \frac{\partial L(\mathbf{W})}{\partial W_{nm}} \end{pmatrix}$$

- From an identical argument we used for vectors

$$L(\mathbf{W} + \epsilon \mathbf{U}) = L(\mathbf{W}) + \epsilon \text{tr} \mathbf{U}^\top \frac{\partial L(\mathbf{W})}{\partial \mathbf{W}} + O(\epsilon^2)$$

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$$\text{Thus } \frac{\partial L(\mathbf{W})}{\partial \mathbf{W}} = 2 (\mathbf{a}^\top \mathbf{W} \mathbf{b} - c) \mathbf{a} \mathbf{b}^\top$$

Traces

- The trace of a matrix is the sum of its diagonal elements

$$\text{tr} \mathbf{A} = \text{tr} \mathbf{A}^T = \sum_i A_{ii}$$

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Quick Matrix Differentiation

- Let

$$\partial_{\mathbf{U}} f(\mathbf{X}) = \lim_{\epsilon \rightarrow 0} \frac{f(\mathbf{X} + \epsilon \mathbf{U}) - f(\mathbf{X})}{\epsilon}$$

- E.g.

$$\begin{aligned} \partial_{\mathbf{U}} \operatorname{tr} \mathbf{A} \mathbf{X} \mathbf{B} &= \lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} (\operatorname{tr} \mathbf{A} (\mathbf{X} + \epsilon \mathbf{U}) \mathbf{B} - \operatorname{tr} \mathbf{A} \mathbf{X} \mathbf{B}) \\ &= \operatorname{tr} \mathbf{A} \mathbf{U} \mathbf{B} \end{aligned}$$

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thus

$$\frac{\partial \text{tr } \mathbf{A} \mathbf{X} \mathbf{B}}{\partial \mathbf{X}} = \mathbf{A}^T \mathbf{B}^T$$

Log Determinants

- We often come across logarithms of determinants of matrices, $\log(|\mathbf{M}|)$
- For GP we want to choose $\mathbf{K}$ to maximise the marginal likelihood, $\log(|\mathbf{K} + \sigma^2 \mathbf{I}|)$
- To find the derivative of $\log(|\mathbf{X}|)$ we consider

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- ★ Using $|\mathbf{AB}| = |\mathbf{A}||\mathbf{B}|$
- ★ Using $\log(ab) = \log(a) + \log(b)$

Determinants

$$|\mathbf{I} + \epsilon \mathbf{M}| = \begin{vmatrix} 1 + \epsilon M_{11} & \epsilon M_{12} \\ \epsilon M_{21} & 1 + \epsilon M_{22} \end{vmatrix}$$

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 \end{aligned}$$

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Determinants

$$|\mathbf{I} + \epsilon \mathbf{M}| = \begin{vmatrix} 1 + \epsilon M_{11} & \epsilon M_{12} \\ \epsilon M_{21} & 1 + \epsilon M_{22} \end{vmatrix} = (1 + \epsilon M_{11})(1 + \epsilon M_{22}) - \epsilon^2 M_{21} M_{12} \\ = 1 + \epsilon(M_{11} + M_{22}) + O(\epsilon^2)$$

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Determinants

$$|\mathbf{I} + \epsilon \mathbf{M}| = \begin{vmatrix} 1 + \epsilon M_{11} & \epsilon M_{12} \\ \epsilon M_{21} & 1 + \epsilon M_{22} \end{vmatrix} = (1 + \epsilon M_{11})(1 + \epsilon M_{22}) - \epsilon^2 M_{21} M_{12} \\ = 1 + \epsilon(M_{11} + M_{22}) + O(\epsilon^2)$$

$$|\mathbf{I} + \epsilon \mathbf{M}| = \begin{vmatrix} 1 & \epsilon M_{12} & \epsilon M_{13} & \epsilon M_{14} & \epsilon M_{15} \\ \epsilon M_{21} & 1 + \epsilon M_{22} & \epsilon M_{23} & \epsilon M_{24} & \epsilon M_{25} \\ \epsilon M_{31} & \epsilon M_{32} & 1 + \epsilon M_{33} & \epsilon M_{34} & \epsilon M_{35} \\ \epsilon M_{41} & \epsilon M_{42} & \epsilon M_{43} & 1 + \epsilon M_{44} & \epsilon M_{45} \\ \epsilon M_{51} & \epsilon M_{52} & \epsilon M_{53} & \epsilon M_{54} & 1 + \epsilon M_{55} \end{vmatrix} \\ = (1 + \epsilon M_{11}) C_{11} - \epsilon M_{21} C_{21} + \epsilon M_{31} C_{31} - \epsilon M_{41} C_{41} + \epsilon M_{51} C_{51}$$

Determinants

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 |\mathbf{I} + \epsilon \mathbf{M}| &= \begin{vmatrix} 1 + \epsilon M_{11} & \epsilon M_{12} \\ \epsilon M_{21} & 1 + \epsilon M_{22} \end{vmatrix} = (1 + \epsilon M_{11})(1 + \epsilon M_{22}) - \epsilon^2 M_{21} M_{12} \\
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 \end{aligned}$$

Determinants

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 &= \prod_i (1 + \epsilon M_{ii}) + O(\epsilon^2)
 \end{aligned}$$

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 |\mathbf{I} + \epsilon \mathbf{M}| &= \begin{vmatrix} 1 + \epsilon M_{11} & \epsilon M_{12} \\ \epsilon M_{21} & 1 + \epsilon M_{22} \end{vmatrix} = (1 + \epsilon M_{11})(1 + \epsilon M_{22}) - \epsilon^2 M_{21} M_{12} \\
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 &= (1 + \epsilon \sum_i M_{ii}) + O(\epsilon^2)
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 &= (1 + \epsilon \operatorname{tr} \mathbf{M}) + O(\epsilon^2)
 \end{aligned}$$

Putting it Together

- Recall

$$\log(|\mathbf{X} + \epsilon \mathbf{U}|) - \log(|\mathbf{X}|) = \log(|\mathbf{I} + \epsilon \mathbf{X}^{-1} \mathbf{U}|)$$

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$$\begin{aligned}\log(|\mathbf{X} + \epsilon \mathbf{U}|) - \log(|\mathbf{X}|) &= \log(|\mathbf{I} + \epsilon \mathbf{X}^{-1} \mathbf{U}|) \\ &= \log(1 + \epsilon \operatorname{tr} \mathbf{X}^{-1} \mathbf{U} + O(\epsilon)^2)\end{aligned}$$

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using $\log(1 + x) = x + \frac{x^2}{2} + \dots$

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using $\log(1 + x) = x + \frac{x^2}{2} + \dots$

- Thus $\partial_{\mathbf{U}} \log(|\mathbf{X}|) = \operatorname{tr} \mathbf{U}^T (\mathbf{X}^{-1})^T$

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- Recall

$$\begin{aligned}\log(|\mathbf{X} + \epsilon \mathbf{U}|) - \log(|\mathbf{X}|) &= \log(|\mathbf{I} + \epsilon \mathbf{X}^{-1} \mathbf{U}|) \\ &= \log(1 + \epsilon \operatorname{tr} \mathbf{X}^{-1} \mathbf{U} + O(\epsilon)^2) \\ &= \epsilon \operatorname{tr} \mathbf{X}^{-1} \mathbf{U} + O(\epsilon)^2 \\ &= \epsilon \operatorname{tr} \mathbf{U}^T (\mathbf{X}^{-1})^T + O(\epsilon)\end{aligned}$$

using $\log(1 + x) = x + \frac{x^2}{2} + \dots$

- Thus $\partial_{\mathbf{U}} \log(|\mathbf{X}|) = \operatorname{tr} \mathbf{U}^T (\mathbf{X}^{-1})^T$

- Or

$$\frac{\partial \log(|\mathbf{X}|)}{\partial \mathbf{X}} = (\mathbf{X}^{-1})^T$$

Summary

- With care you can differentiate most expressions
- The chain and product rule are incredibly powerful tools
- We can generalise differentiation to vectors and matrices
- There are a number of surprisingly useful results

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- There are a number of surprisingly useful results: see **The Matrix Cookbook**
- When we look at **integration** it gets harder